



清华大学物理系
Department of Physics, Tsinghua University

Numerical computation of 2D generalized Brillouin zones

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Bloch band theory

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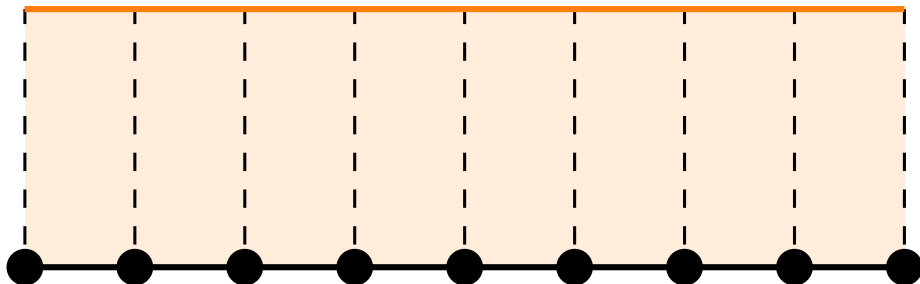
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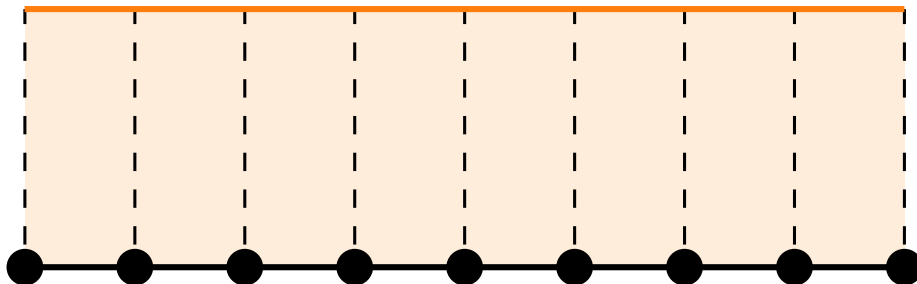


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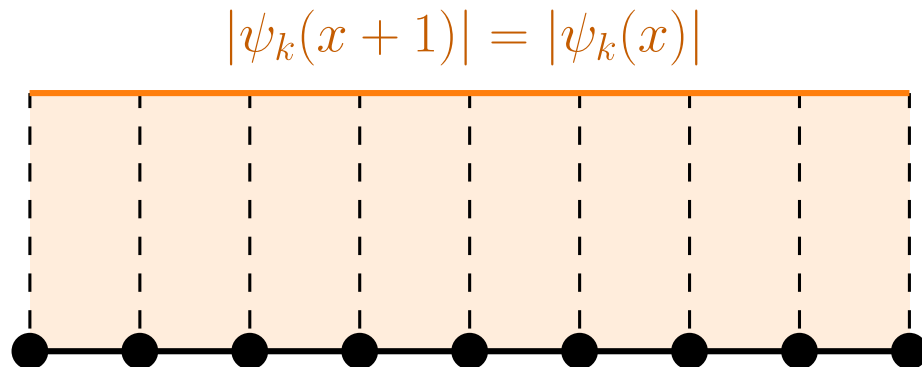


Non-Bloch band theory

- **Non-Hermitian** periodic bulk
 $\psi(x + a) = \beta\psi(x), \beta \in \mathbb{C}, \beta \neq 0$

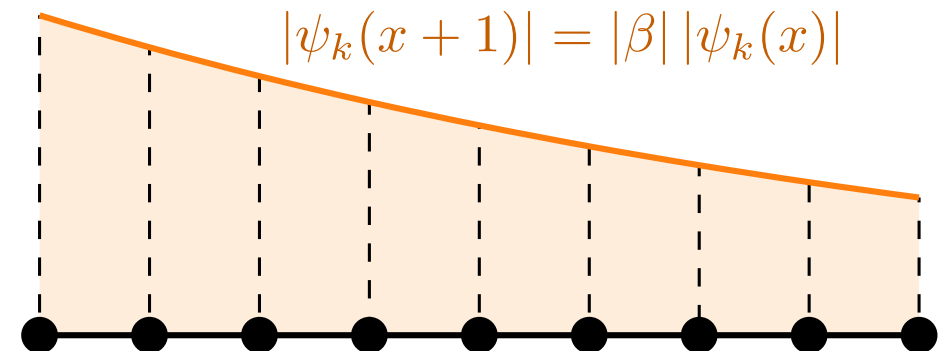
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 $\psi(x + a) = \beta\psi(x), \beta \in \mathbb{C}, \beta \neq 0$
- Bulk spectrum in the OBC limit
 Non-Bloch energy bands $E_{n(\beta)}$
- Bulk OBC states are superpositions of non-Bloch waves $\psi_{\beta,n}(x)$.



- For an n -dimensional lattice, extending $\mathbf{k} \in \mathbb{R}^n \rightarrow \beta \in \mathbb{C}^n$ **doubles the dimensionality** of momentum space.

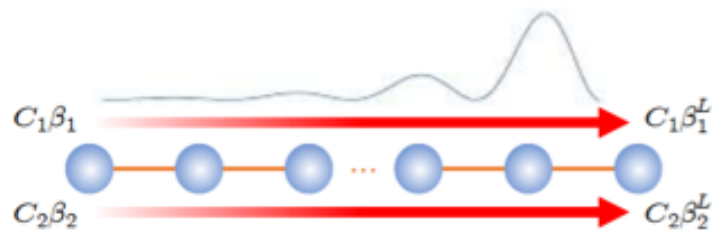
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 - GBZ constraint: $|\beta^{(M)}| = |\beta^{(M+1)}|$



- Yao, S. & Wang, Z.,
Phys. Rev. Lett., 121, 086803 (2018)
- Yokomizo, K. & Murakami, S.,
Phys. Rev. Lett., 123, 066404 (2019)

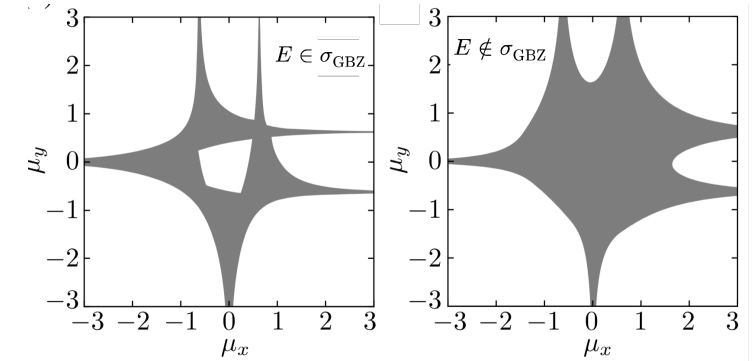
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- Amoeba formulation:

$$E \in \sigma_{\text{Amoeba}} \Leftrightarrow \text{Absence of a "central hole" in } \mathcal{A}(E)$$

- $\mathcal{A}(E) := \{\mu \in \mathbb{R}^n \mid \exists \theta, \det[E - h(e^{\mu+i\theta})] = 0\}$



Wang, H.-Y., Song, F. & Wang, Z.,
Phys. Rev. X, 14, 021011 (2024)

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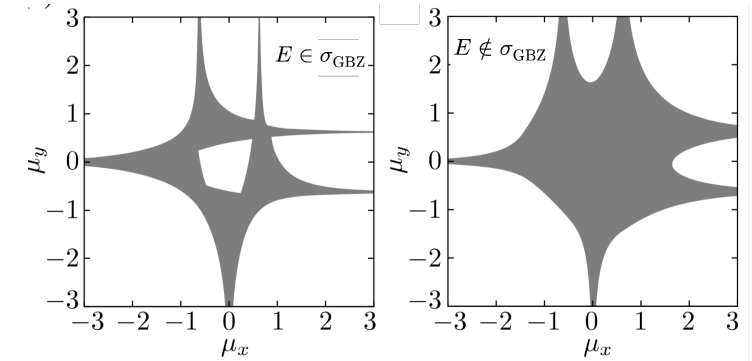
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- Ronkin function:

$$R_E(\boldsymbol{\mu}) = \int_{\mathbb{T}^n} \left(\frac{d\boldsymbol{\theta}}{2\pi} \right)^n \ln |\det[E - h(e^{\boldsymbol{\mu} + i\boldsymbol{\theta}})]|$$

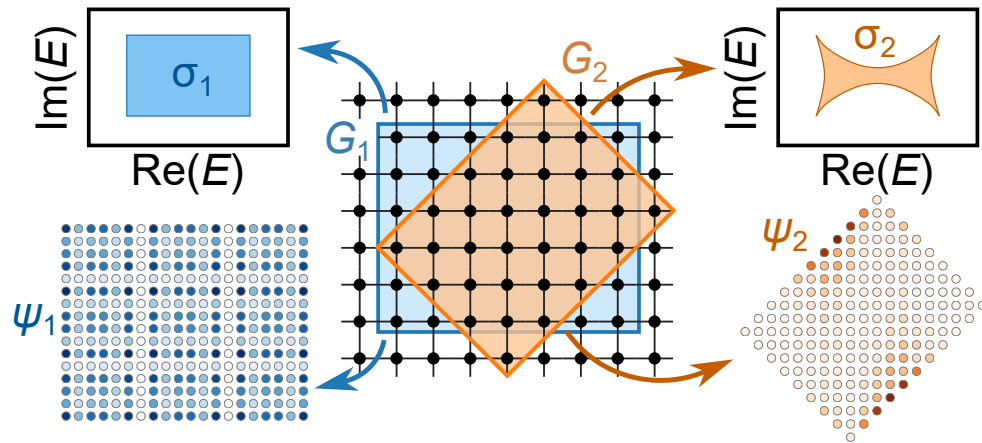
- Central hole: an open complement component where $\nabla R_E = 0$.



Wang, H.-Y., Song, F. & Wang, Z.,
Phys. Rev. X, 14, 021011 (2024)

Geometry-dependent skin effect

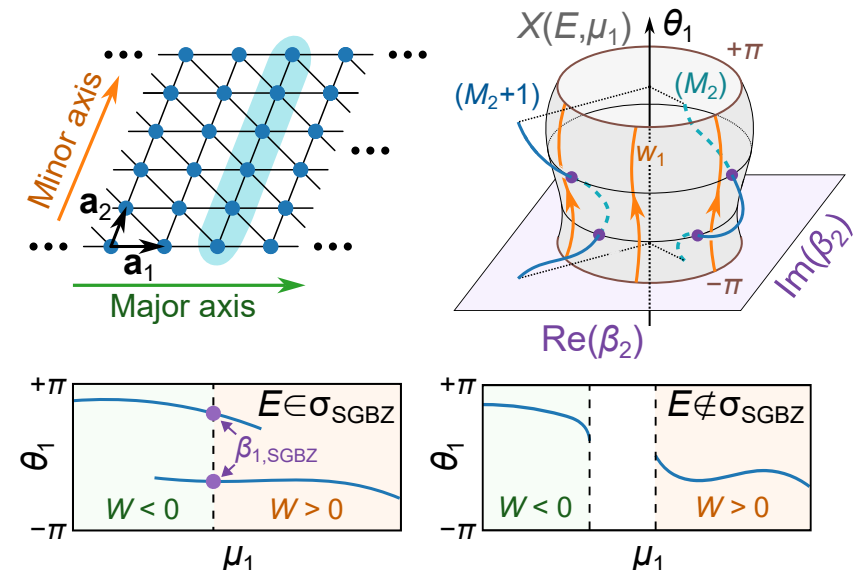
- In $n \geq 2$ dimensions, the OBC spectrum and decay factors depend on boundary geometry.



Zhang, K., et al., *Nat. Commun.*, 13, 2496 (2022)

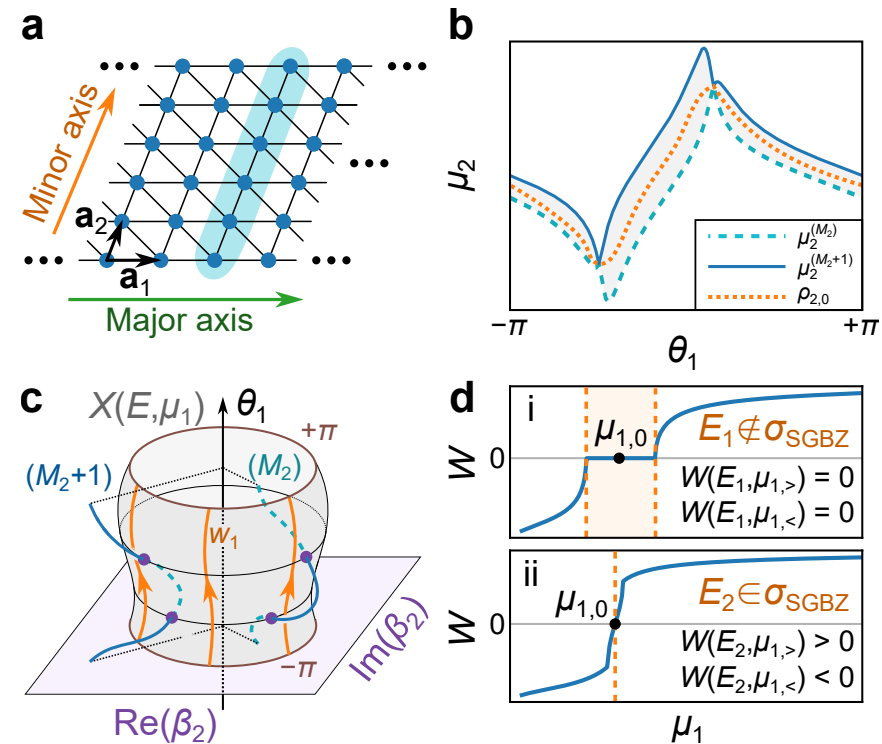
Strip-GBZ formulation

- OBC spectrum in the strip limit ($L_1 \gg L_2$).

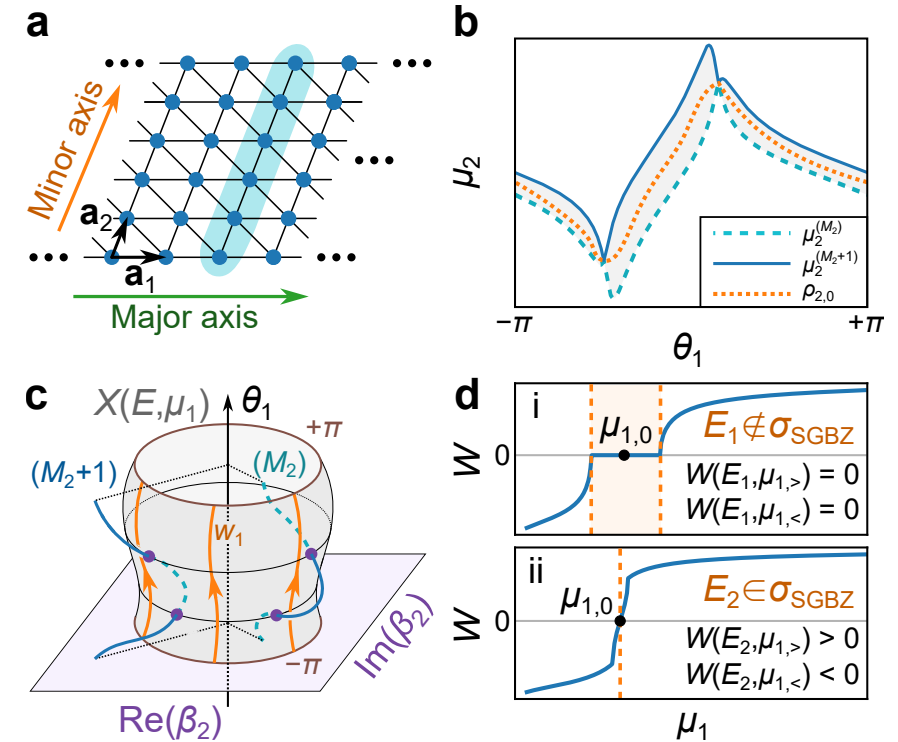


Xiong, Y., et al., *arXiv*, 2407.01296 (2024)

Wang, C., et al., *arXiv*, 2506.22743 (2025)



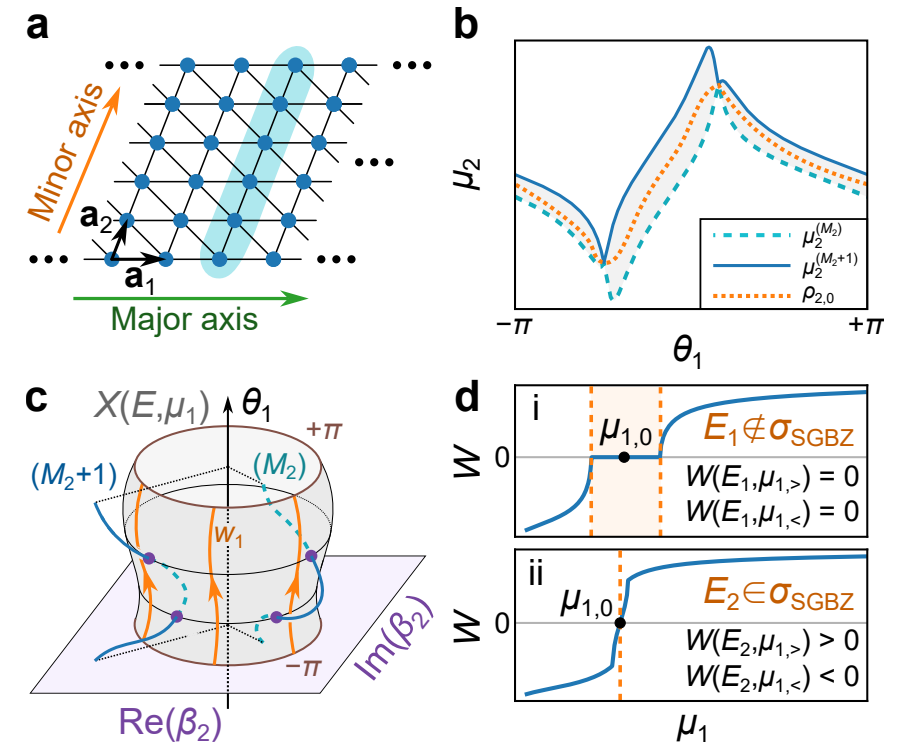
- Minor-axis log-radius function: $\rho_{2,0}(\theta_1; E, \mu_1)$
 Periodic, with $\mu_2^{(M_2)} \leq \rho_{2,0} \leq \mu_2^{(M_2+1)}$



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- Base manifold:

$$X(E, \mu_1) := \left\{ (\beta_1, \beta_2) \mid \theta_1, \theta_2 \in [-\pi, \pi] \right. \\ \left. \beta_1 = e^{\mu_1 + i\theta_1}, \beta_2 = e^{\rho_{2,0}(\theta_1) + i\theta_2} \right\}$$

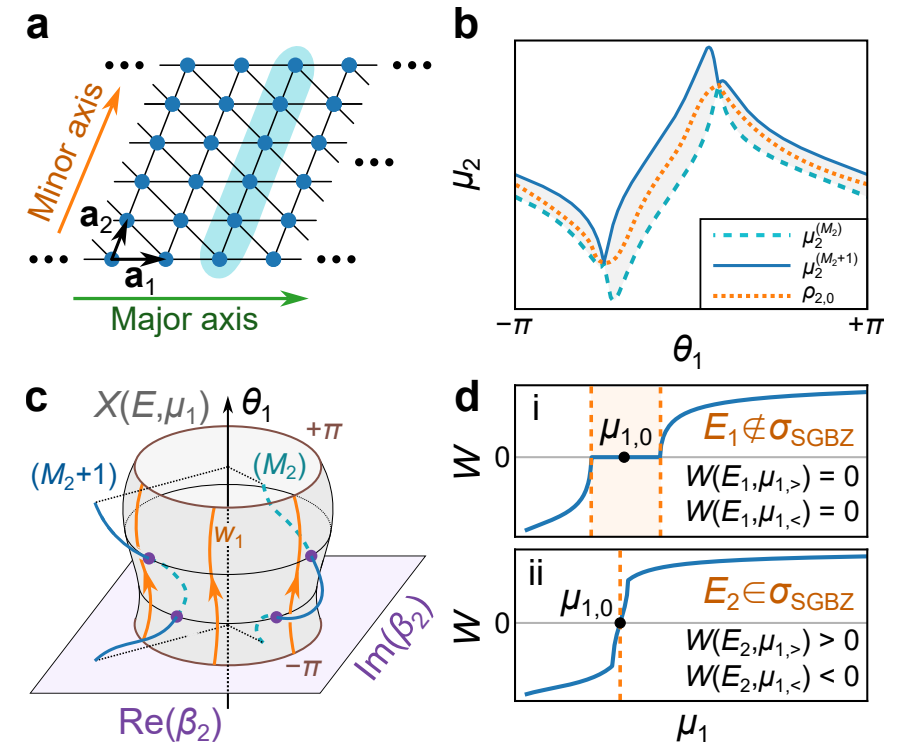


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- Major-axis winding number $w_1(\theta_2; E, \mu_1)$:
Winding of $\det(E - h)$ along θ_1 at fixed θ_2 .



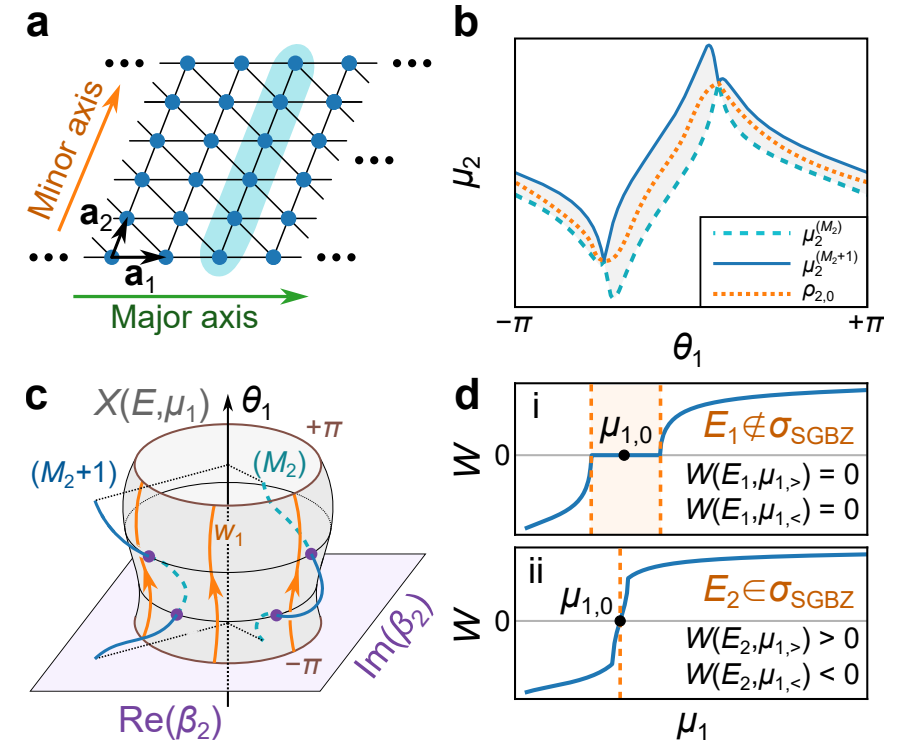
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- Major-axis winding number $w_1(\theta_2; E, \mu_1)$:
Winding of $\det(E - h)$ along θ_1 at fixed θ_2 .
- Average winding number:

$$W(E, \mu_1) := \frac{1}{2\pi} \int_{-\pi}^{\pi} w_1(\theta_2; E, \mu_1) d\theta_2$$

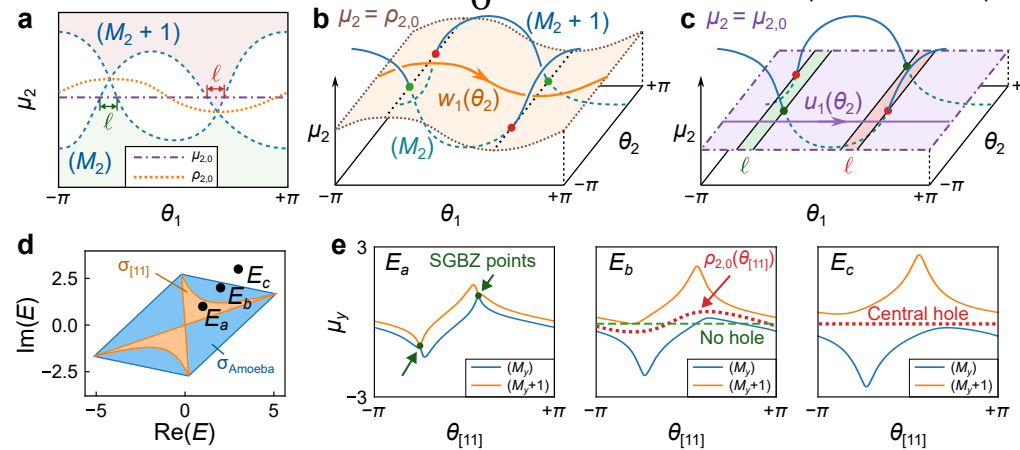


- Derivative of $R_E(\mu) \Leftrightarrow$ average winding number on the torus at fixed μ

$$\frac{\partial R_E(\mu)}{\partial \mu_j} = \int_{\mathbb{T}^{n-1}} \frac{d\theta_1 \cdots d\hat{\theta}_j \cdots d\theta_n}{(2\pi)^{n-1}} u_j(\theta_1, \dots, \hat{\theta}_j, \dots, \theta_n)$$

where

$$u_j(\theta_1, \dots, \hat{\theta}_j, \dots, \theta_n) := \int_0^{2\pi} \frac{d\theta_j}{2\pi i} \frac{\partial_{\theta_j} \det(E - h(e^{\mu+i\theta}))}{\det(E - h(e^{\mu+i\theta}))}$$



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- $\det(E - h(\beta))$: a **continuous** map from the base manifold $\beta \in X$ to the complex plane.
- Projection onto the unit circle: $P : \mathbb{C} \setminus \{0\} \rightarrow S^1, z \mapsto z/|z|$
- Composition:

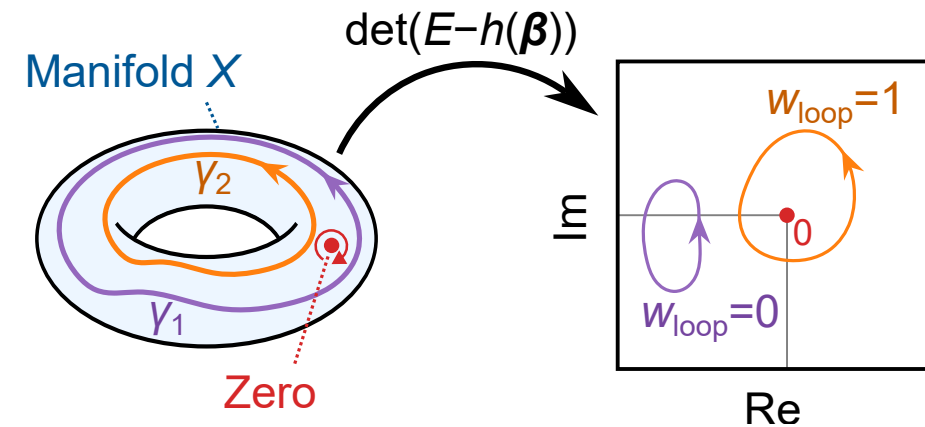
$$X \setminus \{\text{Zeros}\} \xrightarrow{\det[E-h(\beta)]} \mathbb{C} \setminus \{0\} \xrightarrow{P} S^1$$

- Winding number: homomorphism

$$w_{\text{loop}} : \pi_1(X \setminus \{\text{Zeros}\}) \longrightarrow \pi_1(S^1) = \mathbb{Z}$$

- Since $\mathbb{Z}$ is Abelian,

$$w_{\text{loop}} : H_1(X \setminus \{\text{Zeros}\}) \longrightarrow \mathbb{Z}$$



- Strip GBZ: $X = X(E, \mu_1)$
- Amoebic GBZ: $X = X(\mu) := \{(e^{\mu_1 + i\theta_1}, e^{\mu_2 + i\theta_2}) \mid \theta_1, \theta_2 \in [-\pi, \pi]\}$

- Argument principle:

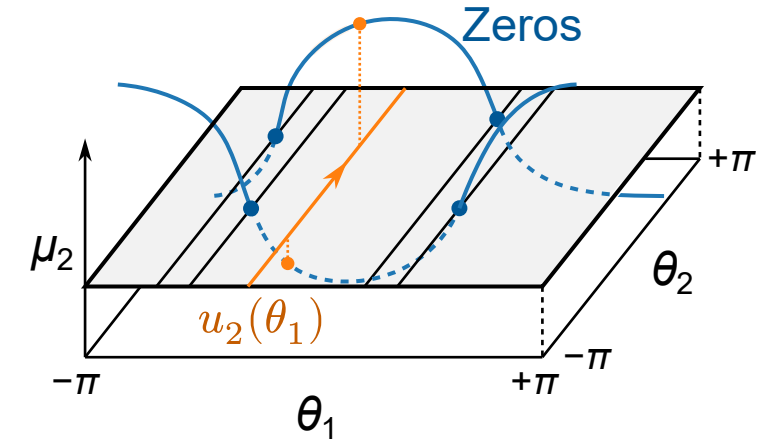
$$\oint_{\mathcal{C}} \frac{dz}{2\pi i} \frac{f'(z)}{f(z)} = N_{\text{zeros}} - N_{\text{poles}}$$

Zeros and poles inside $\mathcal{C}$ count with multiplicity. The contour avoids both.

- Let $-M_j$ and N_j be the lowest and highest powers of β_j in $\det(E - h)$, for $j = 1, 2$.
- Loop winding number:
 - $u_1(\theta_2; \mu_1, \mu_2) = (\# \text{ of } \beta_1 \text{ roots inside } e^{\mu_1}) - M_1$
 - $u_2(\theta_1; \mu_1, \mu_2) = (\# \text{ of } \beta_2 \text{ roots inside } e^{\mu_2}) - M_2$

Algorithm: Compute $\partial_2 R_E(\mu_1, \mu_2)$

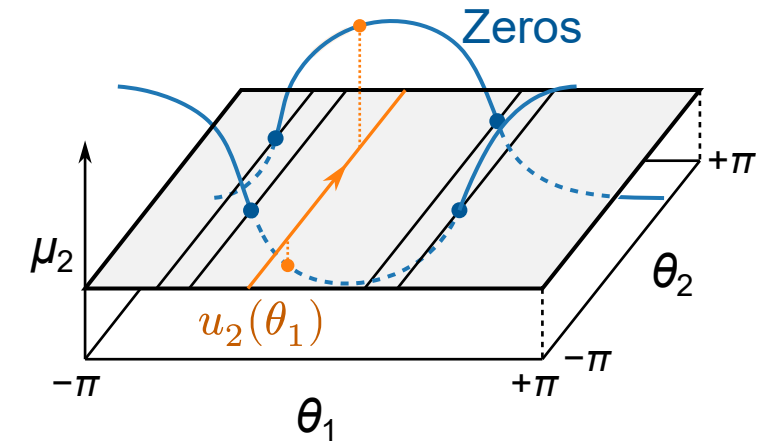
- 1 Compute tracks of roots $(E, \mu_1, \theta_1) \mapsto \beta_2^{(j)}$
 - 2 Collect the intersection points of $\ln|\beta_2^{(j)}| = \mu_2$
 - 3 **if** intersection points are discrete:
 - 4 Partition the θ_1 circle at the crossing angles
 - 5 **For each** interval:
 - 6 Choose the interval midpoint θ_1
 - 7 Count roots with $\ln|\beta_2| < \mu_2$: $u_2 = N - M_2$
 - 8 $\partial_2 R_E = \sum_{\text{intervals}} u_2 \times (\text{interval width})/2\pi$
 - 9 **else**: $\partial_2 R_E$ is ill-defined.
-



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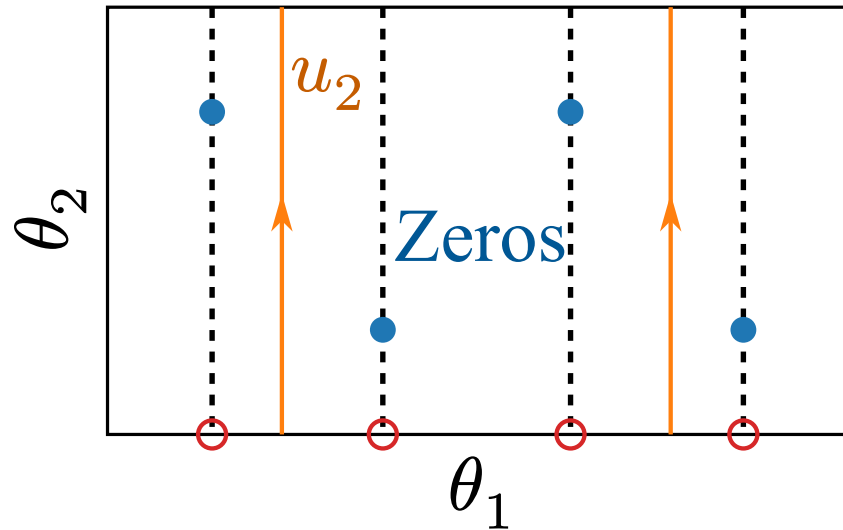
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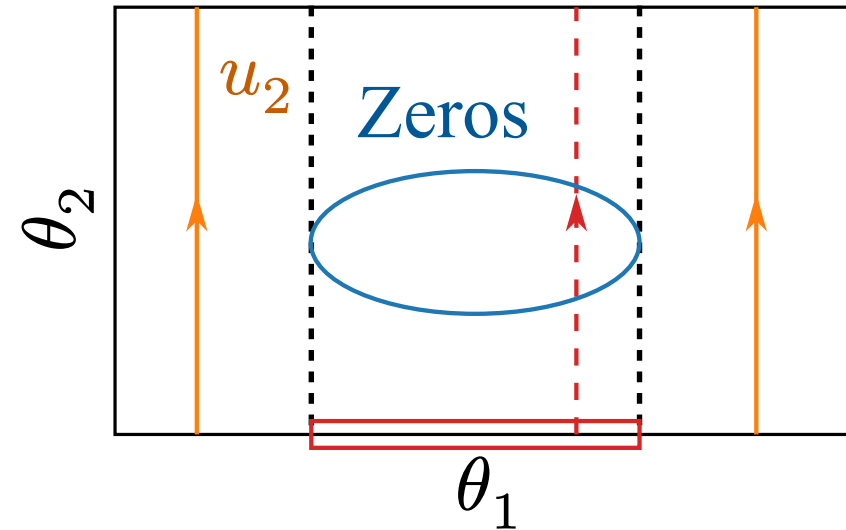
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- $u_2(\theta_1; \mu_1, \mu_2) =$
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- $\partial_1 R_E$ can be computed
 with the same procedure.

Discrete zeros (point subsets)



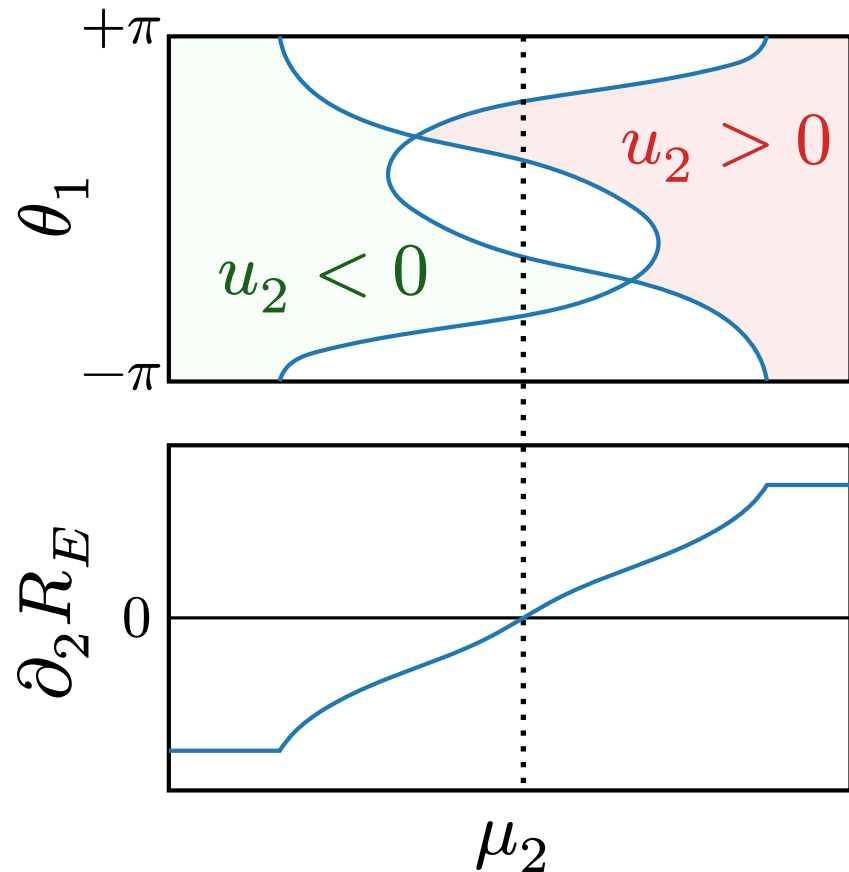
- Singular loop parameters form a set of measure zero.
- $\int_{-\pi}^{\pi} u_2(\theta_1) d\theta_1$ is **well defined**.

Continuum zeros (line subsets)

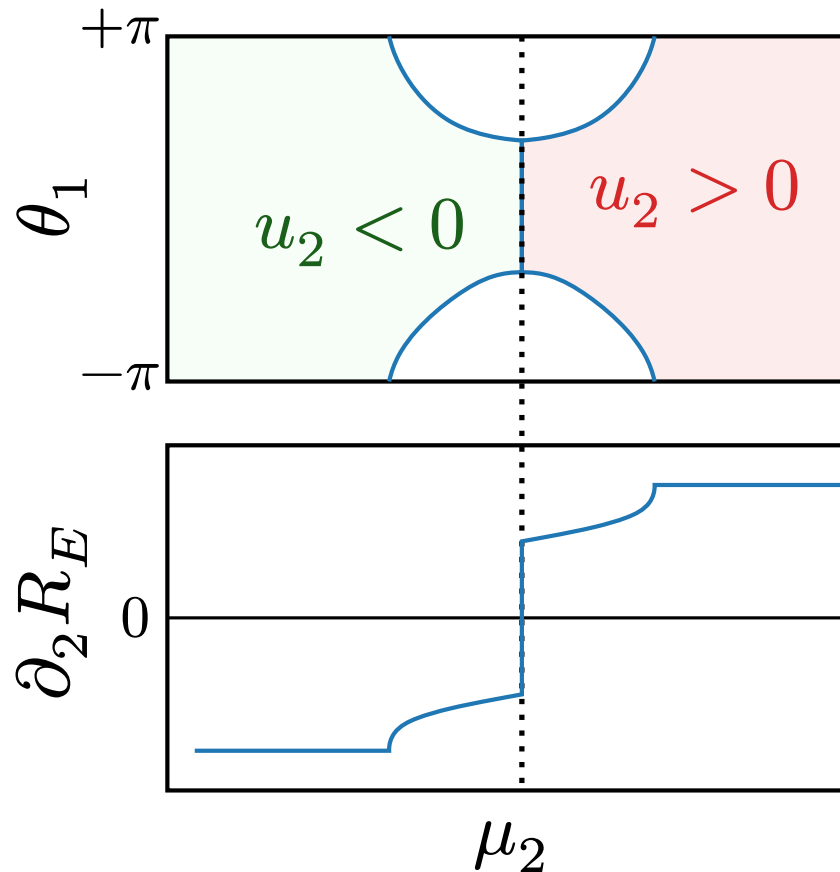


- Singular loop parameters occupy an interval.
- $\int_{-\pi}^{\pi} u_2(\theta_1) d\theta_1$ is **ill-defined**.

Discrete zeros (point subsets)



Continuum zeros (line subsets)



- At fixed E , GBZ subsets satisfy $f(E, \beta_1, \beta_2) = 0$ and $\ln|\beta| = \mu_{\min}$.
- Partial minimization preserves convexity: $F_{E(\mu_1)} := \min_{\mu_2} R_{E(\mu_1, \mu_2)}$ is convex.

Outer search in μ_1

- 1 Bracket the sign change in $\partial_1 R_E$.
 - 2 Choose the midpoint μ_1 .
 - 3 Track $\beta_2^{(j)}(\theta_1)$ once.
 - 4 Run the inner search for μ_2 .
 - 5 Evaluate $\partial_1 R_E$ at that result.
 - 6 Update the μ_1 bracket.
 - 7 Stop at zero winding or a resolved continuum minimum.
-

Inner search in μ_2

- 1 Reuse the tracks at fixed (E, μ_1) .
 - 2 Test the gap and flat-track levels.
 - 3 Bracket the sign change in $\partial_2 R_E$.
 - 4 Locate crossings of $\ln|\beta_2| = \mu_2$.
 - 5 Evaluate $\partial_2 R_E$ by root counts.
 - 6 Bisect until $|\partial_2 R_E| < \varepsilon_w$.
 - 7 Retain refined roots for later levels.
-

- A flat root track at $\ln|\beta_2| = \mu_{2,c}$ makes the usual loop winding singular over an interval of θ_1 .

- Probe the two sides of the level:

$$w_2^- = \partial_2 R_{E(\mu_1, \mu_{2,c}-\varepsilon)}, \quad w_2^+ = \partial_2 R_{E(\mu_1, \mu_{2,c}+\varepsilon)}.$$

- If $w_2^- < 0 < w_2^+$, the level minimizes the convex function in μ_2 . The minimum can occur at a kink where an ordinary derivative does not exist.
- Resolve the outer search with analogous μ_1 probes at fixed $\mu_{2,c}$. Opposite signs identify a continuum minimum and produce line subsets.
- If the signs do not straddle zero, continue on the appropriate side. A zero-winding plateau with no zeros represents an exterior energy.

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- Fix (E, μ_1) and vary θ_1 , with $\beta_1 = \exp(\mu_1 + i\theta_1)$. Solve $f(E, \beta_1, \beta_2) = 0$ for all $K = M_2 + N_2$ roots.
- A uniform θ_1 mesh can underresolve rapid root motion near a multiple root. Sorting roots independently at each step also loses their track identities.
- The continuation state is

$$\mathbf{y} = \left(\theta_1, \ln \beta_2^{(1)}, \dots, \ln \beta_2^{(K)} \right).$$

- Adapt the angular step to motion in this state space. At each new angle, solve the polynomial again and match the roots to their predicted positions.

Implicit differentiation along a regular root gives

$$\partial_{\beta_1} f \cdot i\beta_1 + \partial_{\beta_2} f \cdot \frac{d\beta_2}{d\theta_1} = 0.$$

Hence the logarithmic tangent of track j is

$$V_j := \frac{d \ln \beta_2^{(j)}}{d\theta_1} = -i \frac{\beta_1}{\beta_2^{(j)}} \frac{\partial_{\beta_1} f}{\partial_{\beta_2} f}, \quad \mathbf{V} = (1, V_1, \dots, V_K).$$

The proposed angular increment is

$$\Delta\theta_1 = \frac{h}{\|\mathbf{V}\|_2}, \quad \|\mathbf{V}\|_2 = \sqrt{1 + \sum_j |V_j|^2}.$$

- Fast root motion increases $\|\mathbf{V}\|_2$ and reduces $\Delta\theta_1$.
- h controls the approximate arclength step and adapts to prediction error.
- Undefined tangents of zero/infinite padding roots are excluded from the norm.

1 **Predict:** $\hat{\beta}_2^{(j)} = \beta_2^{(j)} \exp(V_j \Delta \theta_1)$.

2 **Solve:** compute all roots $\tilde{\beta}_2^{(k)}$ at the trial angle.

Match: find a one-to-one assignment π minimizing

3
$$\sum_j d_{\text{chord}}\left(\hat{\beta}_2^{(j)}, \tilde{\beta}_2^{(\pi(j))}\right).$$

4 **Accept or retry:** compare matched roots with the predictions.

- Hungarian matching uses distance on the Riemann sphere:

$$d_{\text{chord}}(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}.$$

- The sphere metric handles roots near zero and infinity. A prediction anchor reduces track swaps at close approaches.
- Each regular root sample comes from a polynomial solve.

Use the largest matched prediction error:

$$e = \frac{\max_j d_{\text{chord}} \left(\hat{\beta}_2^{(j)}, \tilde{\beta}_2^{(\pi(j))} \right)}{\text{atol} + \text{rtol} \cdot \text{median}_{k:\text{finite}} |\tilde{\beta}_2^{(k)}|}.$$

Accepted step: $e < 1$

- Store the solved roots in track order.
- Propose $h_{\text{next}} = h \min(10, 0.9e^{-\frac{1}{2}})$.
- For $e = 0$, use the maximum growth factor.

Rejected step: $e \geq 1$

- Reduce h and solve again.
- Use $h \leftarrow h \max(0.2, 0.9e^{-\frac{1}{2}})$.
- Stop after the retry budget or minimum step is reached.

Defaults: $\text{atol} = 10^{-12}$, $\text{rtol} = 10^{-3}$, $h_{\text{max}} = 0.5$. An accepted step cannot grow immediately after a rejection.

Trigger	Candidate condition
Step collapse	$\Delta\theta_1 < 10^{-10}$ near a singular implicit derivative.
Pairwise approach	For a nearby root pair, $d \beta_i - \beta_j ^2/d\theta_1$ changes from negative to positive.

- Refine a point candidate by solving $f = 0$ and $\partial_{\beta_2} f = 0$. For an interval candidate, first locate its distance-derivative zero.
- A close approach is only a candidate. Confirm a finite-root cluster before splitting the segment.
- Store the multiple root as a shared endpoint, restart beyond it, and track the next segment.
- At $\theta_1 = 2\pi$, retain the permutation relating the closing roots to $\theta_1 = 0$.

- ZeroManager returns tracked roots, analytic tangents, and multiple-root boundaries on an adaptive mesh.
- **SGBZ**: refine equal-modulus events, build $\mu_{2,\text{mid}}(\theta_1)$, and extract boundary points or line intervals.
- **Amoeba**: locate crossings of $\ln|\beta_2| = \mu_2$ and reuse the same tracks throughout the inner μ_2 search.
- Cubic Hermite interpolation supplies matching anchors and the SGBZ loop path. Singular endpoint derivatives require linear fallback.
- Step control resolves root motion. Additional crossing and extremum refinement resolves the level sets used by each solver.

CONTENTS



清華大學 物理系

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- Introduction to 2D generalized Brillouin zones
- Computing average winding via polynomial root-solving
- Pseudo-arclength continuation of zeros
- Benchmark: 2D complex Hatano-Nelson model



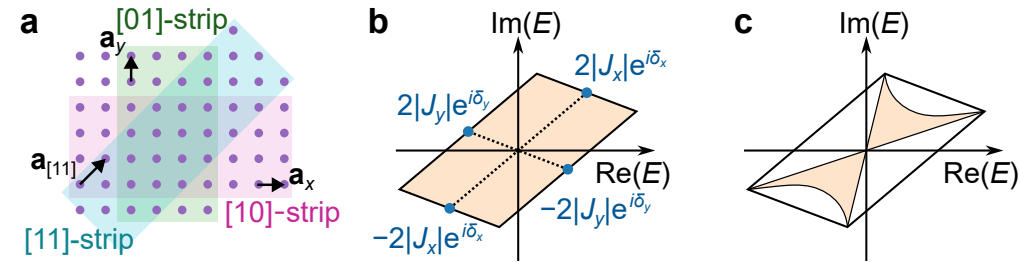
$$h(\beta_x, \beta_y) = J_{x1}\beta_x^{-1} + J_{x2}\beta_x + J_{y1}\beta_y^{-1} + J_{y2}\beta_y.$$

For $\alpha = x, y$, write the nonzero hoppings as

$$J_{\alpha 1} = \exp(\gamma_\alpha + i\delta_\alpha)J_\alpha,$$

$$J_{\alpha 2} = \exp(-\gamma_\alpha + i\delta_\alpha)J_\alpha^*.$$

- γ_α : nonreciprocal decay rate.
- δ_α : common hopping phase.
- Compare amoeba with x -, y -, and $[11]$ -strip GBZs.



Wang, C., et al., *arXiv*, 2506.22743v3, Eqs. (S3.16), (S3.30) (2025)

Cartesian coordinates: amoeba, x strip, and y strip

$$|\beta_x| = \exp(\gamma_x), \quad |\beta_y| = \exp(\gamma_y).$$

Diagonal strip: $(\beta_1, \beta_2) = (\beta_x \beta_y, \beta_y)$

$$|\beta_1| = \exp(\gamma_x + \gamma_y), \quad A = \frac{J_{x1}}{\beta_1} + J_{y2}, \quad B = J_{x2}\beta_1 + J_{y1}.$$

At fixed β_1 , the characteristic equation becomes

$$A\beta_2^2 - E\beta_2 + B = 0, \quad |\beta_2| = \sqrt{\left| \frac{B}{A} \right|}.$$

- The product of the two roots is $\frac{B}{A}$. Equal moduli give the transverse GBZ radius.
- The [11] radius generally varies with θ_1 . These formulas apply where A and B are nonzero.

For every returned point and every stored line sample, compare

$$\varepsilon_{\mu} = \max_{j=1,2} \max_n |\ln|\beta_{j,n}| - \mu_j^{\text{exact}}(\theta_{1,n})|,$$

$$\varepsilon_E = \max_n \frac{|E - h(\beta_{x,n}, \beta_{y,n})|}{|E| + \left|\frac{J_{x1}}{\beta_{x,n}}\right| + |J_{x2}\beta_{x,n}| + \left|\frac{J_{y1}}{\beta_{y,n}}\right| + |J_{y2}\beta_{y,n}|}.$$

Case	$(J_{x1}, J_{x2}, J_{y1}, J_{y2})$	E
Point subsets	$(1 + i, 1.5 + 1.2i, -1 + i, -1.2 - 0.5i)$	$1 + i$
Line subsets	$(1, 1.5, -1, -1.2)$	1

- Acceptance thresholds: $\varepsilon_{\mu} \leq 10^{-5}$ and $\varepsilon_E \leq 10^{-7}$.
- Check spectral membership independently, including successful empty results.

Case	GBZ	Samples	ϵ_{μ}	ϵ_E
Points	Amoeba	4	9.52×10^{-9}	2.01×10^{-16}
Points	x strip	4	2.00×10^{-9}	1.00×10^{-16}
Points	y strip	4	1.44×10^{-9}	1.00×10^{-16}
Points	[11] strip	4	1.44×10^{-9}	2.10×10^{-16}
Lines	Amoeba	354	5.97×10^{-7}	1.32×10^{-10}
Lines	x strip	350	4.22×10^{-7}	1.32×10^{-10}
Lines	y strip	346	5.80×10^{-7}	1.28×10^{-10}
Lines	[11] strip	380	3.98×10^{-7}	5.51×10^{-11}

All eight cases pass. Each point case contains four points and each line case contains two line subsets.

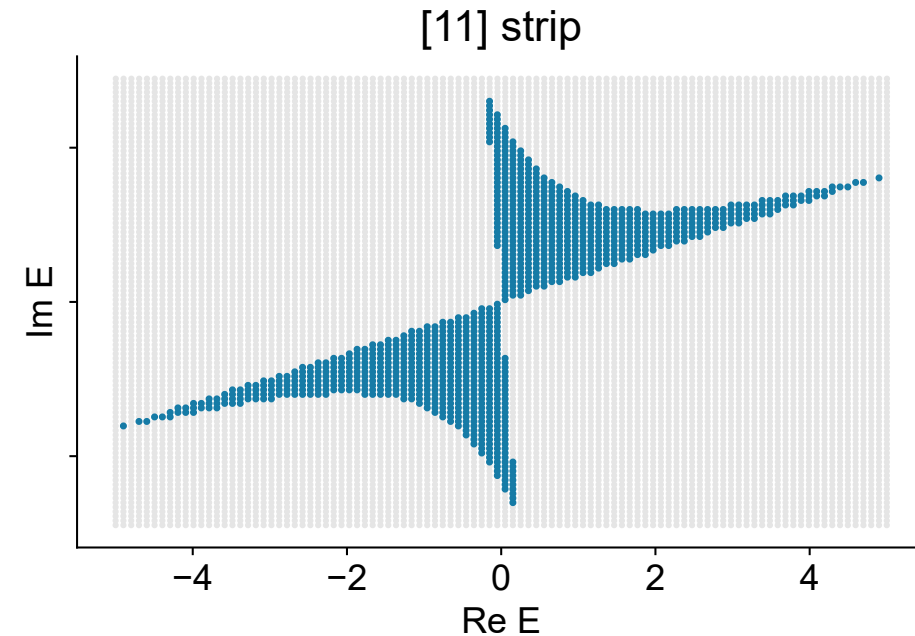
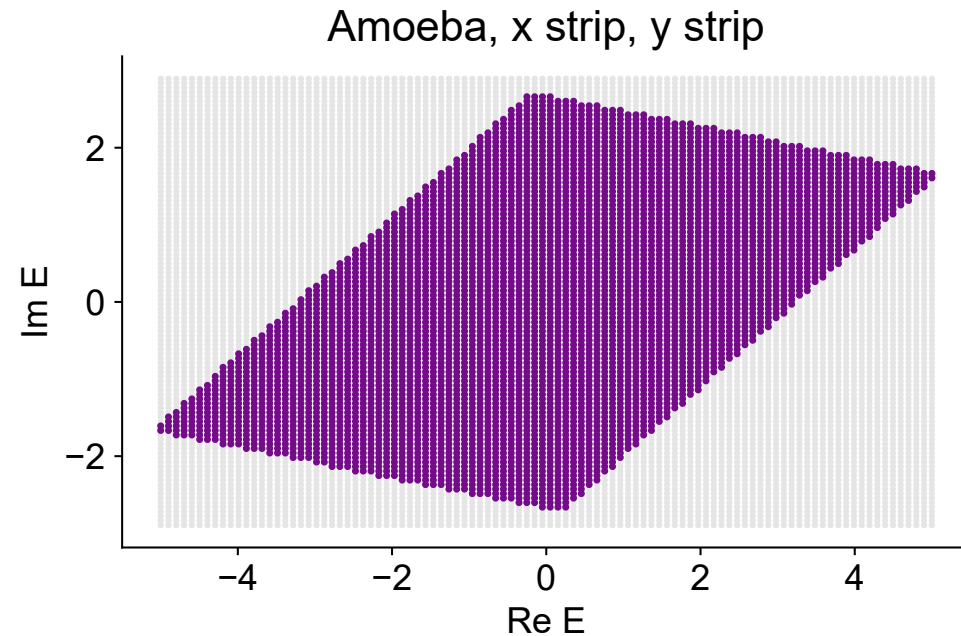
NumPy backend, default solver settings. Sample accuracy does not by itself prove that every branch was found.

Let $u = 2 \exp(\mathrm{i}\delta_x)|J_x|$ and $v = 2 \exp(\mathrm{i}\delta_y)|J_y|$.

- Cartesian spectra: $E \in \{us + vt : s, t \in [-1, 1]\}$.
- Diagonal strip: $E^2 \in \text{conv}\{0, (u + v)^2, (u - v)^2\}$.

Hoppings	E	Amoeba / x / y	[11] strip
Complex case	$6 + 6\mathrm{i}$	Outside	Outside
Real case	6	Outside	Outside
Real case	i	Outside	Outside
Complex case	$2 + 2\mathrm{i}$	Inside	Outside

Sixteen solver comparisons agree with the independent analytic classification: 13 empty results and 3 nonempty results. Failed solves do not count as exterior energies.



- All four methods evaluate the same 100×100 grid, with zero failed results.
- Amoeba, x strip, and y strip share all 4,832 in-spectrum grid points. The [11] strip contains 1,108 points on the same grid.

Complex hopping case. Dots are computed grid points. Finite sampling does not certify arbitrarily narrow spectral features.